What course should I study? Investigating the application of a Recommender System for Irish college courses

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Supervisor: Owen Conlan

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Declaration

I, the undersigned, declare that this work has not previously been submitted as an exercise for a degree at this, or any other University, and that unless otherwise stated, is my own work.

Brendan McCann

February 26, 2026

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...ABSTRACT... **TODO**

Acknowledgments

TODO

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Chapter 1

Introduction

1.1 Motivation

This section describes the context behind known problems in the research domain and provides some background knowledge. In light of these established problems, a technology is introduced, one that could be appropriately applied to this domain. The applicability of the proposed solution is also discussed, all relevant issues in describing the research motivation behind this dissertation.

According to Higher Education Authority (2024), Department of Education (2024), 79% of Irish secondary school students progressed to college after graduation. The students who have considered progressing to college have all been posed with the question: *What course should I study?* This is a daunting question for many reasons. The majority of secondary school students are faced with making this decision at 16-19 years old, an age with limited life experience and knowledge about the world. In Smyth et al. (2011), Irish students were reported to experience significantly increasing levels of stress in their final year of secondary school, complicating this decision further. This decision also comes with many personally confronting questions, which may include: *"What do I want to do with the rest of my life?"* and *"What do I love doing?"*, among many others. Moreover, there exists the fear of choosing a college course the student regrets - in Higher Education Authority (2018), 31% of recent Irish college graduates reported they were not likely to study their college course again, adding more weight to this crucial life decision.

Choosing a college course is not only a challenging decision to wrestle with as an individual. Students are shown to be influenced by a multitude of external factors when choosing their college course. For instance in Aydın (2015), parents are shown to be one of the most influential factors of a student's college course choice. Aydın (2015) have also shown that an individual's gender, socioeconomic status and their parent's educational background are among the most prominent societal influences of an individual's college course choice, adding complexity to a decision that is already challenging.

To assist students in this difficult decision, it is worth introducing the role of a Guidance Counsellor (GC). GC's are trained professional in Irish secondary schools, primarily known for assisting students in their career prospects after graduation. It is worth noting that the role of a GC extends well beyond vocational support - according to Institute of Guidance Counsellors (2016), GC's are also trained to provide educational, personal and social support. In this context, it is worth introducing the Institute of Guidance Counsellors (IGC) [1], the professional body for GC's in Ireland. In Institute of Guidance Counsellors (2022), mental health issues were more commonly encountered by GC's compared to career decision-making issues [2], marking the first time that career decision-making issues were not the most commonly encountered issue by GC's since 2019. In the IGC's 2024-2027 strategic plan outlined in Institute of Guidance Counsellors (2024b), the IGC noted this stronger prevalence of mental health issues compared to career decision-making issues, reflecting their change in strategy for GC's going forward, signalling a shift in priorities for GC's across Irish secondary schools: vocational support may not be the most important aspect of a GC's role going forward.

We can shift our focus to the more technical side of things. A Recommender System (RS) is a technology that recommend items to a user. Well-known examples of RS's include Netflix [3] recommending films to their customers, or Amazon [4] suggesting products to their customers. One of the earliest known RS's was *Tapestry* in Goldberg et al. (1992), a system that made tailored mailing list recommendations to its users. Around this time, the field of RS's received considerable interest as e-commerce websites could increase their revenue if they suggested more personalised products to their users. As discussed in Schafer et al. (1999), Amazon and Netflix are the most notable adopters of RS's in order to increase company revenue. There are a wide variety of approaches to recommending items to users, although, two very common types of RS's are *Collaborative Filtering* and *Content-based Filtering* RS's. Following the definitions in Aggarwal et al. (2016), *Collaborative Filtering* recommends items to users based on what other similar users also liked, whereas *Content-based Filtering* recommends items to users based on the

[1] `https://igc.ie/`

[2] In Institute of Guidance Counsellors (2024a), career decision-making issues did surpass mental health issues as the most commonly encountered issue for GC's, however, both are pressing issues.

[3] `https://www.netflix.com/`

[4] `https://www.amazon.com/`

user's preferences.

As previously explained, an RS recommends *items* to a *user*. However, we can re-conceptualize an RS in the context of college courses: *college courses* can take the place of *items*, and a *student* can take the place of a *user*. Given the established difficulty for an individual choosing a suitable college course - including internal preferences and external influences, the shifting priorities of GC's towards student's mental health as opposed to career-making decisions, and a technology that could be appropriately applied to this domain, an exciting opportunity presents itself: A Recommender System for Irish College Courses.

A technology, such as an RS, could aid a student in navigating a difficult decision-making process. For instance, there are over 45 higher education institutes in Ireland offering a total of over 1700 courses [5], representing a huge set of choice to the student. An advantage of a personalised RS is to provide a user with a much smaller list of relevant recommendations. Long et al. (2025) demonstrated that smaller recommendation sets are often equally as effective as larger recommendations sets, suggesting that a much narrower set of personalised college course recommendations could be effective in this space.

It is worth noting that there are no intentions behind this RS to replace the vocational support role of a GC. Choosing a college course is a difficult and personally confronting decision, and as mentioned in Institute of Guidance Counsellors (2016), GC's are trained to additionally support a student's social and personal needs, allowing the GC to provide a student with empathetic and human-centered support - an aspect that a traditional RS would fail to effectively convey.

Furthermore, there are no intentions behind this RS to entirely replace the student's college course decision-making process. As we will discover in Chapter 3, the decision of a college course is incredibly nuanced. It is the author's opinion that no system can or should make that decision entirely for an individual. The intention behind this RS is to *complement*, rather than replace, the student's college course decision-making process, assisting the student in exploring a variety of career paths that may be relevant to them.

This ultimately describes the motivation behind this research.

[5]Higher education colleges and courses can be found here: `https://careersportal.ie/`

1.2 Research Question

In light of the motivation behind this research, the Research Question for this dissertation can be formulated: *How can a Recommender System assist an individual in their college course decision-making process?*

1.3 Research Objectives

With the motivation and research question formulated, the research objectives for this dissertation are outlined below:

1. Conduct thorough background research into an individual's influences and motivations behind choosing a college course.
2. Identify the relevant information to question a user in order to make suitable college course recommendations.
3. Create an Irish college course RS with statistically significant improvements compared to a baseline college course RS.
4. Analyse user evaluation metrics gathered from a conducted experiment.
5. Explore the qualities that would make an effective college course RS.

1.4 Overview

In this section, an overview of this dissertation's chapters are provided.

- Chapter 2- Background Reading. This chapter describes the necessary background knowledge to understand the Irish secondary school and college admission system.
- Chapter 3 - Literature Review. This chapter provides the reader with a solid understanding of the nuances behind an individual's college course choice. There will be explorations into an individual's motivations behind choosing a college course, the external influences of an individual's college course choice, and the relevant information to question a user in order to make suitable college course recommendations, among other relevant discussions. As well as this, there will be an exploration into existing implementations of college course RS's, both in literature and real-world products. This includes a discussion into their successes and limitations.
- Chapter 4 - Initial Prototypes. **TODO**.

- Chapter 5 - Design. **TODO**
- Chapter 6 - Implementation. **TODO**
- Chapter 7 - Methodology. This chapter describes the experiment methodology in which the RS was tested with human participants.
- Chapter 8 - Evaluation. This chapter provides an objective analysis of the experiment evaluation metrics, and provides a qualitative analysis of verbal feedback gathered from the experiment.
- Chapter 9 - Conclusions & Future Work. **TODO**

Chapter 2

Background Reading

As mentioned in Chapter 1, this dissertation specifically focuses on a College Course RS in the context of the Irish secondary school and college system. Grading systems and college admission systems often differ between countries, so this chapter seeks to provide essential background knowledge behind the Irish secondary school and the college admission system. Section 2.1 explains the Leaving Certificate, Section 2.2 describes the National Framework of Qualifications, Section 2.3 explains the Central Applications Office, Section 2.4 describes Irish college entry requirements, and finally a summary of the chapter is provided in Section 2.5. Upon completion of the chapter, the reader should be familiar with the workings of the Irish secondary school and college admission system.

2.1 The Leaving Certificate

The Leaving Certificate (LC) is the final examination taken by the majority of secondary school students in Ireland. Instead of taking the LC, students can take the Leaving Certificate Applied (LCA) which is a more practical alternative to the LC. Students who take the LCA cannot directly enter higher education institutes; they commonly go for direct employment, apprenticeships or enter further education institutes. For the sake of narrowing scope, apprenticeships and further education courses are considered beyond the scope of this RS. The RS proposed in this dissertation will only recommend higher education college courses.

A student's overall performance in the LC is measured in terms of points, a number that ranges between 0 and 625, where 625 is the maximum achievable points. A student chooses to be examined in each subject at increasing levels of difficulty, known as Higher

Level (HL), Ordinary Level (OL) or Foundation Level (FL) [1]. A student receives a grade depending on the level and mark they achieve in the subject. For example, a student that gets a mark of 91% in the HL LC English exam receives a H1 grade, for which they receive 100 points, and a student that gets a mark of 64% in their OL LC Geography exam receives an O4 grade, for which they receive 28 points [2]. A student is examined in a minimum of 6 subjects, and their overall LC points is an accumulation of the points received in each examined subject.

2.2 NFQ Levels

The Irish National Framework of Qualifications (NFQ) [3] is a system that describes all education qualifications in Ireland. It is a 10-level system, where 1 is the lowest and 10 is the highest (doctorate degree). For instance, the LC is classed as NFQ Level 5. The NFQ levels that are relevant to this disseration are NFQ Level 6 (Advanced or Higher Certificate), NFQ Level 7 (Ordinary Bachelors degree), NFQ Level 8 (Honours Bachelor degree) and NFQ Level 9 (Masters degree).

2.3 The Central Applications Office

Rather than going to the colleges directly to apply for a specific course, in Ireland all college applications are done through the Central Applications Office (CAO) [4], the centralised college admissions system in Ireland. As part of a CAO application, students are encouraged to create two numbered lists, each with up to 10 college courses. One of these numbered lists are for NFQ Level 8 college courses, and the other list is for NFQ Level 6 and 7 college courses. These college courses are ordered by the student's preference of study - for instance, the student's number one choice would be the course they want to study the most, and their number ten choice is the course they want to study the least out of this list.

The preference ranking of a student's CAO application courses is relevant here. For example, if a student receives an offer for a course that is ranked first in their list, they will not receive offers for courses below this preference, even if they satisfy the course's entry requirements. Entry requirements will be explained in section 2.4.

[1] Foundation Level is only offered for the LC Irish and Mathematics subjects.

[2] For a full list of LC exam grades and their translated LC points, see `https://careersportal.ie/leaving-cert-results-whats-next/calculate-your-leaving-cert-points`.

[3] `https://www.qqi.ie/what-we-do/the-qualifications-system/national-framework-of-qualifications`

[4] `https://www.cao.ie/`

2.4 Entry Requirements

Entry requirements for colleges differ by country, and in this section the entry requirements for Irish college courses is described.

2.4.1 LC Points

The primary entry requirement for most college courses in Ireland is LC points. For example, the Computer Science (CS) course in Trinity College Dublin (TCD) has a minimum entry requirement of 533 points [5], meaning that a student must achieve 533 points or higher in their LC to be considered admission into the course. A course's LC points is the points of the student who was last admitted to the course in the previous academic year. For example, let's suppose that in 2025, 150 people were admitted into CS in TCD. Out of all these 150 students, the student with the lowest LC points was 533, meaning the course has a minimum entry requirement of 533 LC points. As a result, the LC points associated with a course are not fixed and fluctuate every year. LC points are commonly misunderstood as reflecting the difficulty of a course, however, it would be more accurate to say that the LC points reflect the *demand* of a course.

2.4.2 LC Subject Requirements

Many Irish college courses have grade requirements for specific subjects to allow admission into the course. In most cases, the subjects are relevant to the course and the college seeks knowledge in that subject. For example, CS in TCD has a minimum entry requirement of a H4 (60-69% in HL) in Mathematics to be considered for admission.

In Ireland, students are obligated to take the LC English, Mathematics and Irish [6] exams, and they choose a minimum of four additional subjects to take exams in. The student makes their subject choices in fifth year, which is the penultimate year in Irish secondary schools. Due to the differing subject entry requirements by college courses, some students may be unable to take certain courses simply because they have not taken the relevant subject. For example, if a student has not taken two science [7] subjects, they are unable to be admitted into Medicine [8], meaning that a student's college course options are already limited by the subject choices they made in fifth year, almost two years before

[5] 533 points as of 2025, see: `https://www.tcd.ie/courses/undergraduate/courses/computer-science/`

[6] Some students may qualify for an exemption from Irish if they satisfy certain requirements.

[7] LC Physics, Chemistry, Physics/Chemistry, Biology and Agricultural Science.

[8] Subject requirements for TCD Medicine as of 2026: `https://www.tcd.ie/courses/undergraduate/courses/medicine/`

they finalise their CAO college course list.

In addition to specific college courses requiring a minimum grade in particular subjects, some subjects are required to be admitted into the college itself. For example, all Royal College of Surgeons in Ireland (RCSI) courses [9] require a third language [10].

2.4.3 Portfolios, Tests or Interviews

In a majority of cases, if a student satisfies the minimum entry requirements for a course with their LC points and subject grades, the student is admitted into the course. However, some Irish college courses have additional entry requirements. For instance, some courses may require the student to be examined in a test outside of the Leaving Certificate. For example, all Medicine courses in Ireland require their applicants to sit the Health Professions Admissions Test (HPAT) [11]. Some college courses may also require a portfolio submission, or for the student to undertake an interview. For example, the Architecture course in Technological University Dublin (TUD) [12] requires the student to submit an additional portfolio showcasing the student's competence in sketching, painting, etc., as well as an interview demonstrating their interest in the course. Some courses may also require students to go through Garda Vetting [13], a background check for individuals working with children or vulnerable adults, an example of this includes General Nursing in TCD [14].

In the case of Portfolios and Tests, the students score in these are added to their total LC points. For example, if a student achieves 550 LC points and a score of 200 in the HPAT, their combined LC points is 750 only when the student is applying for courses that require the HPAT. This explains why some courses may have a minimum requirement of 732 LC points, despite 625 being the maximum achievable LC points.

2.5 Summary

In summary, this chapter provided the reader with knowledge surrounding the workings of the Irish secondary school and college admissions system. In particular, the LC and LC points system was explained, including the grading system and the way in which subjects are examined at different levels - for example, Higher Level or Ordinary Level. The reader

[9] As of 2026, see: `https://www.rcsi.com/dublin/undergraduate/medicine/entry-requirements`
[10] Langauges other than English and Irish, for example: German, Spanish, Italian, etc.
[11] `https://hpat-ireland.acer.org/`
[12] `https://www.tudublin.ie/study/undergraduate/courses/architecture-tu832/`
[13] `https://vetting.garda.ie/`
[14] `https://www.tcd.ie/courses/undergraduate/courses/nursing---general-nursing/`

was introduced to the workings of NFQ Levels and the CAO system, developing an understanding into the Irish college admission system. In addition to this, the varying entry requirements into specific courses or colleges were explained. With this chapter complete, the reader should have developed a broad understanding behind the Irish secondary school and college admission system.

Chapter 3

Literature Review

In this chapter, the reader will be provided with a solid understanding of the nuances behind an individual's college course choice. A discussion of an individual's factors in choosing a college course is provided in Section 3.1. Section 3.2 seeks to explore the external and societal influences on an individual's college course choice. Section 3.4 pays a visit to the field of Vocational Psychology and examines the theories of college and career choice. Similarly, Section 3.5 examines the theories of job and career satisfaction. Interviews with expert guidance counsellors were conducted as part of this research, and a qualitative analysis of these interviews are provided in Section 3.6. Finally, a summary of this chapter is provided in Section 2.5. After reading this chapter, the reader should have a broad understanding of the complexities an individual faces when choosing a suitable college course.

3.1 Individual factors

3.2 External influences

3.3 College influences

3.4 Theories of career choice & college course choice

3.5 Theories of job satisfaction & career satisfaction

3.6 Interviews with guidance counsellors

Chapter 4

Initial Prototypes

Chapter 5

Design

At the beginning of each chapter, a description should introduce the reader to the content of the chapter. The description should explain to the reader the layout of the chapter, the contribution that the chapter makes to the overall dissertation and the contribution of the individual sections towards the overall chapter.

5.1 Problem Formulation

This section should provide the reader with an overall description of the problem that will be addressed in the dissertation. In contrast to a generic discussion of the dissertation topic in the Introduction chapter, this section should provide a detailed discussion of the problem that has been identified based on the existing work that has been discussed in the preceeding chapter.

In some dissertations, it may make sense to convert this section into a short chapter of its own which follows the discussion of the existing work and preceeds the discussion of the work of the dissertation.

5.1.1 Identified Challenges

This section should present a short description of the gaps in the existing work and the relationship of these gaps to the work described in this dissertation.

5.1.2 Proposed Work

This section should provide a thorough description of the problem and an overview of the work proposed to address the problem.

5.2 Overview of the Design

A description of the approach that addresses the problem identified above.

5.3 Summary

Every chapter aside from the first and last chapter should conclude with a summary.

Chapter 6

Implementation

Guess what? At the beginning of each chapter, a description should introduce the reader to the content of the chapter. The description should explain to the reader the layout of the chapter, the contribution that the chapter makes to the overall dissertation and the contribution of the individual sections towards the overall chapter.

6.1 Overview of the Solution

The code in listing **??** is a demonstration how to include a file with code into the template.

6.2 Component One

The code in listing 6.1 is a demonstration how to include code in the template.

```
x = 1
if x == 1:
    # indented four spaces
    print("x is 1.")
```

Listing 6.1: Second Lengthy caption

6.3 Summary

Every chapter aside from the first and last chapter should conclude with a summary.

Chapter 7

Methodology

Chapter 8

Evaluation

The Evaluation chapter should present a comparison of the work that forms the basis of the dissertation and existing work. At a higher level, it should demonstrate an awareness of the relationship of the dissertation work to the research area that it is based in.

8.1 Experiments

In the case where experiments have been carried out, the experimental setup and the values that were defined for the variables need to be presented in a table e.g. table 8.1.

Column 1	**Column 2**	**Column 3**
Row 1	Item 1	Item 2
Row 2	Item 1	Item 2
Row 3	Item 1	Item 2
Row 4	Item 1	Item 2

Table 8.1: Caption that explains the table to the reader

8.2 Results

Figures that present results such as figure **??** need to display descriptions of the axes, the units and scales of the measurements, statistical values, etc. Where measurements were taken from experiments, error bars or confidence intervals need to be provided to give the reader an indication of the spread of the measurements.

8.3 Summary

Every chapter aside from the first and last chapter should conclude with a summary that presents the outcome of the chapter in a short, accessible form.

Chapter 9

Conclusions & Future Work

This chapter should summarize the work presented in the dissertation and discuss the conclusions that can be drawn from the work and the results presented in chapter **??**.

9.1 Future Work

The section may present a list of items that were beyond the scope of the dissertation.

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